Reductions and NP-completeness

Exercise sheet 9 (hints)

Exercise 9.1. Recall that $L^* = \{w_1 \cdots w_r \mid r \geq 0,\ w_i \in L\}$. Suppose $L \in \text{NP}$. Is it true that $L^* \in \text{NP}$? Prove your answer.

Hint: A certificate can record the cuts and a certificate for each factor.

Exercise 9.2. Prove that INDEPENDENT-SET is NP-complete, where

Input: A finite undirected graph G and a non-negative integer k encoded in binary.

Question: Does G contain a set of at least k nodes with no edge between any two of them?

Hint: What happens to a clique when edges and non-edges are exchanged?

Exercise 9.3. Prove that VERTEX-COVER is NP-complete, where

Input: A finite undirected graph G and a non-negative integer k encoded in binary.

Question: Does G have a set of at most k nodes that contains at least one endpoint of every edge?

Hint: If a set covers every edge, what can you say about the nodes outside it?

Exercise 9.4. Prove that HITTING-SET is NP-complete, where

Input: A finite set U and a family $S_1, \ldots, S_m \subseteq U$, with the elements and sets explicitly listed, and a non-negative integer k encoded in binary.

Question: Is there a set $H \subseteq U$ with $|H| \leq k$ such that $H \cap S_i \neq \emptyset$ for every $i \in \{1, \ldots, m\}$?

Hint: Reduce from VERTEX-COVER.

Exercise 9.5. Prove that DOUBLE-SAT is NP-complete, where

Input: A Boolean formula φ, encoded as a word.

Question: Does φ have at least two distinct satisfying assignments?

Hint: Can you double the number of possible assignments?

Exercise 9.6. Prove that 3SAT is NP-complete, where

Input: A CNF formula with exactly three literal occurrences in every clause. A literal may occur more than once in a clause.

Question: Is the formula satisfiable?

Hint: Start with a clause of four literals. Split it into two clauses using a fresh variable, once positively and once negatively. Extend the idea to longer clauses.

Exercise 9.7. * Prove that SUBSET-SUM is NP-complete, where

Input: A list of non-negative integers $a_1, \ldots, a_m$ and a non-negative integer t. All integers are encoded in binary.

Question: Is there an index set $I \subseteq \{1, \ldots, m\}$ such that

$$\sum_{i \in I} a_i = t?$$

Hint: Reduce from 3SAT. *Let φ have variables $x_1, \ldots, x_n$ and clauses $C_1, \ldots, C_m$. Work with numbers that have $n + m$ decimal digits.*

Exercise 9.8. * Prove that KNAPSACK is NP-complete, where

Input: A list of pairs $(w_1, v_1), \ldots, (w_n, v_n)$ of non-negative integers, and non-negative integers W and V. All integers are encoded in binary.

Question: Is there an index set $I \subseteq \{1, \ldots, n\}$ such that

$$\sum_{i \in I} w_i \leq W \qquad \text{and} \qquad \sum_{i \in I} v_i \geq V?$$

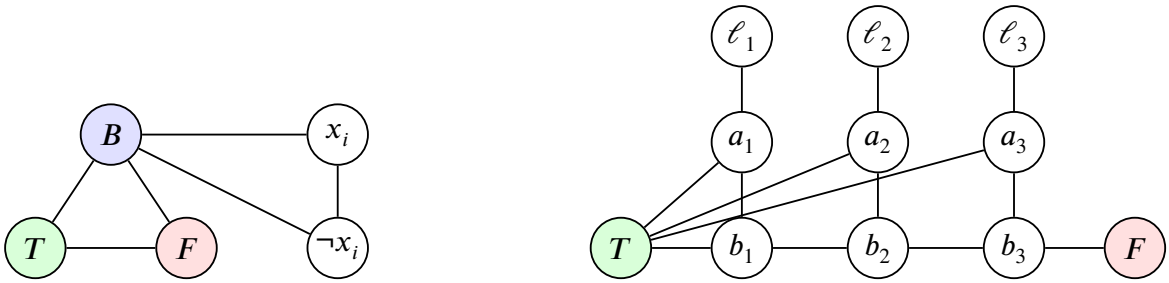
Hint: Reduce from SUBSET-SUM.

Exercise 9.9. * Prove that 3COLOUR is NP-complete, where

Input: A finite undirected graph G.

Question: Can the nodes of G be coloured with at most three colours so that the endpoints of every edge have different colours?

Hint: Reduce from 3SAT. *The gadget on the left sets up the three colours T, F, B and encodes each variable using the nodes x_i and $\neg x_i$. What colours can this pair receive?*



The gadget on the right is for a clause with literal nodes ℓ_1, ℓ_2, ℓ_3 and six fresh nodes a_i, b_i. Its T and F nodes are the same palette nodes as on the left. Suppose each literal node is already in either T or F. What does the gadget on the right enforce?

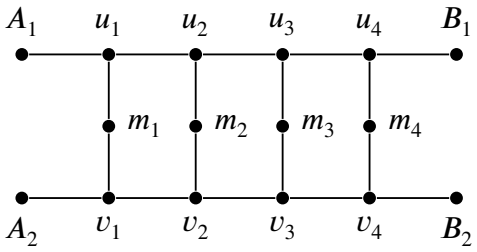
Exercise 9.10. * Prove that HAMILTONIAN-CYCLE is NP-complete, where

Input: A finite undirected graph G.

Question: Does G have a cycle that visits every node exactly once before returning to its starting node?

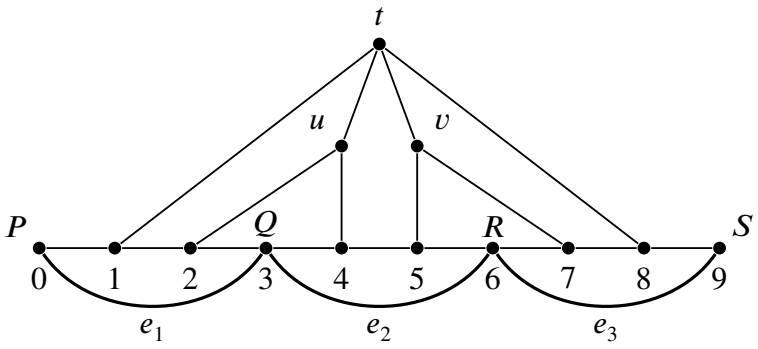
Hint: Reduce from 3SAT. *Here are two gadgets.*

The first has twelve internal nodes; A_1, B_1, A_2, B_2 are outside nodes. No other connections to the outside are allowed.



Suppose this is a part of a bigger graph, connected to it via the external nodes A_i and B_i. What are the ways the Hamiltonian cycle could traverse it?

The second gadget has three distinguished thick edges e_1, e_2, e_3.



Look for a path from $P = 0$ to $S = 9$ visiting all thirteen nodes. Which subsets of the three thick edges can such a path use?

Exercise 9.11. * A nondeterministic Turing machine may have several possible transitions for a state and observed symbol. It accepts an input if some computation halts in the accepting state, and rejects if all computations reject. It runs in polynomial time if every computation halts within a polynomial number of steps.

Prove that $L \in \mathrm{NP}$ if and only if a polynomial-time nondeterministic Turing machine decides L.

Hint: In one direction, let the machine guess a certificate and then run the verifier. How do you bound every branch? In the other direction, what information lets a deterministic verifier replay one chosen computation? The machine has a fixed finite number of transition choices.

Exercise 9.12. * Let NEXP be the class of languages decided by exponential-time nondeterministic Turing machines. Prove that if $\mathrm{P} = \mathrm{NP}$, then $\mathrm{EXP} = \mathrm{NEXP}$.

Hint: Consider a computation together with a time bound written in unary. Checking a proposed accepting computation is then polynomial in this longer input. Under the assumption, what deterministic algorithm becomes available? Apply it to an exponentially large time bound, carefully distinguishing the original input length from the padded length.